The Chinese University of Hong Kong (Shenzhen)
Advanced Math for AI
Fall 2026

Assignment 3

AIE6001 **Total: 100 points + 10 bonus points**

Due: Thursday, 29 October 2026, 23:59 (UTC+8)

Instructions. Show your reasoning and all essential steps. The required questions are worth 100 points in total. Problem 3(d) is optional and offers 10 additional bonus points, for a maximum score of 110.

1. Variance **(12 points)**

In class we defined the variance of a real-valued random variable X as the expected squared deviation from its mean:

$$\mathrm{Var}(X) := \mathbb{E}\big[(X-\mu)^2\big],\ \mu = \mathbb{E}[X]. \tag{1}$$

(a) Show that an equivalent definition for variance is $\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2$. (4 points)

(b) For data points $x_1, \ldots, x_m \in \mathbb{R}$, the empirical variance is $\widehat{\mathrm{Var}} := \frac{1}{m}\sum_{i=1}^m (x_i - \bar{x})^2$ with $\bar{x} = \frac{1}{m}\sum_{i=1}^m x_i$. Show that the empirical variance can be computed from all m^2 ordered pairwise squared differences: (8 points)

$$\widehat{\mathrm{Var}} = \frac{1}{2m^2}\sum_{i=1}^m\sum_{j=1}^m (x_i - x_j)^2$$

2. Independence and Correlation **(25 points)**

Recall that two random variables X and Y with a joint probability mass function or density are independent if their joint function factors as

$$p_{X,Y}(x,y) = p_X(x)p_Y(y). \tag{2}$$

Assume throughout that X and Y have finite, positive variances.

(a) Show that independence implies that X and Y are uncorrelated, i.e., (5 points)

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} = 0.$$

(b) The converse does not hold in general. Let $X \sim \mathrm{Uniform}[-1,1]$ and $Y = |X|$. Show that X and Y are uncorrelated but not independent. (10 points)

(c) Suppose that X and Y are jointly Gaussian. Show that they are independent if and only if they are uncorrelated. *Hint:* Show that when $\rho = 0$, the bivariate Gaussian density factors as in (2). (10 points)

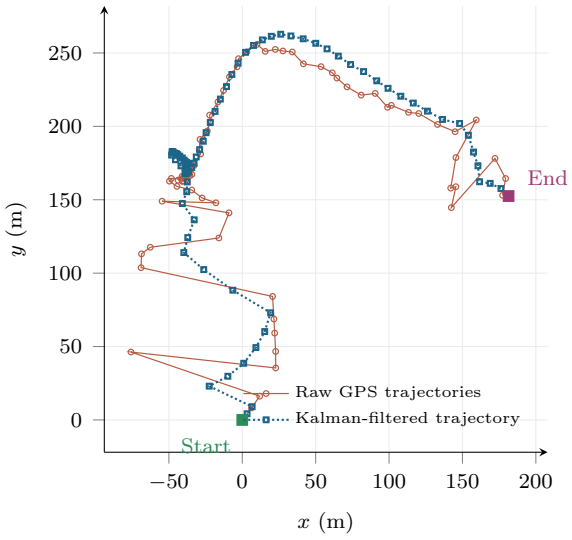
3. Conditional Gaussian Distributions (30 points + 10 bonus)

Consider the time-series model

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(0, Q),$$
$$\mathbf{y}_t = C\mathbf{x}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(0, R).$$

Here $\mathbf{x}_t \in \mathbb{R}^d$ is an unobserved state and $\mathbf{y}_t \in \mathbb{R}^k$ is an observation. Coefficient matrices $A \in \mathbb{R}^{d\times d}$ and $C \in \mathbb{R}^{k\times d}$ describe the state evolution and observation, with $\mathbf{w}_t$ and $\boldsymbol{\varepsilon}_t$ representing additive Gaussian white noise for the state transition and observation, respectively. Assume that all noise vectors are mutually independent and independent of $\mathbf{x}_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$. The objective is to determine $p(\mathbf{x}_t \mid \mathbf{y}_{1:t})$ recursively, where $\mathbf{y}_{1:t} = (\mathbf{y}_1, \ldots, \mathbf{y}_t)$.

For example, we can use this model to track a vehicle moving in a plane with noisy GPS measurements (e.g., see Figure 1). Let the state be $\mathbf{x}_t = (x_t, y_t, \dot{x}_t, \dot{y}_t)^\top$, where x_t, y_t denote position and $\dot{x}_t, \dot{y}_t$ denote velocity. Under a constant-velocity model with sampling interval Δt, $A = \left(\begin{smallmatrix} I_2 & \Delta t\, I_2 \\ 0 & I_2 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} I_2 & 0 \end{smallmatrix}\right)$. Thus A encodes $\Delta x = \dot{x}_t \Delta t$ and $\Delta y = \dot{y}_t \Delta t$, while C restricts the observations to the positions x_t and y_t. Here $\mathbf{w}_t$ accounts for deviations from constant velocity, and $\boldsymbol{\varepsilon}_t$ accounts for GPS error. Thus $p(\mathbf{x}_t \mid \mathbf{y}_{1:t})$ gives us the filtered (denoised) position of the vehicle given past observations.



(a) Show that $(\mathbf{x}_0, \ldots, \mathbf{x}_T)$ is jointly Gaussian. (10 points)

(b) Suppose that $p(\mathbf{x}_t \mid \mathbf{y}_{1:t}) = \mathcal{N}(\mu_t, \Sigma_t)$. Show that $p(\mathbf{x}_{t+1} \mid \mathbf{y}_{1:t}) = \mathcal{N}(m, P)$, where $m = A\mu_t$ and $P = A\Sigma_t A^\top + Q$. (8 points)

(c) Let $S = CPC^\top + R$. Show that $(\mathbf{x}_{t+1}, \mathbf{y}_{t+1}) \mid \mathbf{y}_{1:t}$ is jointly Gaussian with mean $\left(\begin{smallmatrix} m \\ Cm \end{smallmatrix}\right)$ and covariance $\left(\begin{smallmatrix} P & PC^\top \\ CP & S \end{smallmatrix}\right)$. (12 points)

(d) (**Optional**) Assume that S is invertible. For a new observation $\mathbf{y}_{t+1} = \hat{\mathbf{y}}$, show that $p(\mathbf{x}_{t+1} \mid \mathbf{y}_{1:t}, \mathbf{y}_{t+1} = \hat{\mathbf{y}}) = \mathcal{N}(\mu_{t+1}, \Sigma_{t+1})$, where $\mu_{t+1} = m + K(\hat{\mathbf{y}} - Cm)$, $\Sigma_{t+1} = P - KCP$, and $K = PC^\top S^{-1}$. *Hint:* Apply the conditional-distribution formula for a jointly Gaussian vector. (10 bonus)

Remark. The matrix K is the *Kalman gain*. Parts (b) and (d) give the prediction and update equations of the Kalman filter.

Properties of Absorption. For a finite Markov chain that is absorbed with probability one:

1. Let a_i be the probability of reaching absorbing state s from state i. Then $a_s = 1$, $a_i = 0$ at other absorbing states, and $a_i = \sum_{j \in \mathcal{S}} p_{ij} a_j$ at transient states.
2. Let μ_i be the expected time to absorption from state i. Then $\mu_i = 0$ at absorbing states and $\mu_i = 1 + \sum_{j \in \mathcal{S}} p_{ij} \mu_j$ at transient states.

4. Absorbing Markov Chains **(25 points)**

Players M and N have a combined capital of \$3. In each game, the loser pays the winner \$1. M wins each game with probability 2/3. Play continues until one player is bankrupt. The objective is to determine M's probability of winning and the expected number of games played.

(a) Let the state X_t represent M's current amount of money after t games. Draw the Markov-chain transition graph and write the transition matrix P, where $p_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$. (5 points)

(b) Classify each state as transient or absorbing. Show that play ends with probability one. (5 points)

(c) Let h_i be the probability that M eventually wins when starting with \$$i$ (i.e., $X_0 = i$). What are h_0 and h_3? By conditioning on the outcome of the first game, derive equations relating h_1 and h_2. *Hint:* Apply Property 1 on absorption. (4 points)

(d) Suppose M starts with \$1 and N with \$2. Determine M's probability of winning and compare it with 2/3, the probability that M wins an individual game. *Hint:* Solve the equations in part (c). (4 points)

(e) Let m_i be the expected number of games until one player is bankrupt when M starts with \$$i$. What are m_0 and m_1? *Hint:* Apply Property 2 on absorption. (7 points)

Remark. This problem is known as *Gambler's Ruin* and appeared in Pascal and Fermat's correspondence in 1656. It is a classical model of how initial capital and the probability of winning each game affect the risk of bankruptcy.

5. Gaussian Moment Generating Function **(8 points)**

Let $X \sim \mathcal{N}(\mu, \sigma^2)$, where $\sigma > 0$. The moment generating function of X is defined by $M_X(t) = \mathbb{E}[e^{tX}]$. Prove that $M_X(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$ for all $t \in \mathbb{R}$ through the following steps.

(a) Complete the square to show that (6 points)

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu - \sigma^2 t)^2}{2\sigma^2} \right) dx.$$

(b) Complete the proof. *Hint:* A Gaussian density integrates to one. (2 points)